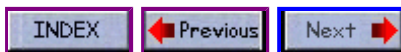


C Programming



FILE INPUT/OUTPUT

To work with files, the library routines must be included into your programs. This is done by the statement,

```
#include <stdio.h>
```

as the first statement of your program.

USING FILES

- **Declare a variable of type FILE**

To use files in C programs, you must declare a file variable to use. This variable must be of type **FILE**, and be declared as a pointer type.

FILE is a predefined type. You declare a variable of this type as

```
FILE *in_file;
```

This declares *in_file* to be a pointer to a file.

- **Associate the variable with a file using fopen()**

Before using the variable, it is associated with a specific file by using the *fopen()* function, which accepts the pathname for the file and the access mode (like reading or writing).

```
in_file = fopen( "myfile.dat", "r" );
```

In this example, the file **myfile.dat** in the current directory is opened for **read** access.

- **Process the data in the file**

Use the appropriate file routines to process the data

- **When finished processing the file, close it**

Use the *fclose()* function to close the file.

```
fclose( in_file );
```

The following illustrates the *fopen* function, and adds testing to see if the file was opened successfully.

```
#include <stdio.h>
/* declares pointers to an input file, and the fopen function */
FILE *input_file, *fopen ();
```

```
/* the pointer of the input file is assigned the value returned from the
/* fopen tries to open a file called datain for read only. Note that */
/* "w" = write, and "a" = append. */
input_file = fopen("datain", "r");

/* The pointer is now checked. If the file was opened, it will point to t
/* character of the file. If not, it will contain a NULL or 0. */
if( input_file == NULL ) {
    printf("*** datain could not be opened.\n");
    printf("returning to dos.\n");
    exit(1);
}
```

NOTE: Consider the following statement, which combines the opening of the file and its test to see if it was successfully opened into a single statement.

```
if(( input_file = fopen ("datain", "r" )) == NULL ) {
    printf("*** datain could not be opened.\n");
    printf("returning to dos.\n");
    exit(1);
}
```

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